

COMP 1101 - Final

Olcay Taner YILDIZ, Farshad MIRAMIRKHANI, Onder UMUT, Bora AKSAR

1. Which of the following codes print numbers from 1 to 10?

a)

```
for i = 0 : 9
    disp("i ");
end
```

b)

```
for i = 0 : 9
    disp(i);
end
```

c)

```
for i = 1 : 10
    disp("i ");
end
```

d)

```
for i = 1 : 10
    disp(i);
end
```

e) None of the above

2. What will be the output of the following program?

```
total = 0;
for i = 1 : 10
    for j = 10 : -1 : i
        total = total + 1;
    end
end
disp(total);
```

a) 55

b) 210

c) 110

d) 50

e) None of the above

3. What will be the output of the following program?

```
total = 0;
for i = 1 : 5
    for j = i : 5
        for k = j : 5
            total = total + 1;
        end
    end
end
disp(total);
```

a) 55

b) 35

c) 210

d) 50

e) 110

4. What will be the output of the following program?

```
displayed = "";
for i = 1 : 3
    for j = 1 : 3
        displayed = displayed + "*";
    end
end
disp(displayed);
```

a) ***

b) ***

**

*

c) *

**

d) *****

e) None of the above

5. Consider the following incomplete code:

```
function result = f(number)
    // Missing body
end
disp(f(3));
```

The missing method body could be

a) result = "number"

b) result = number

b) disp(number)

b) disp("number")

e) None of the above

6. Let say you have the functions factorial and power

```
function result = factorial(N)
function result = power(x, y)
```

which calculate $N!$ and x^y respectively. Which of the following code fragments then calculate the following sum:

$$\frac{1!}{2^1} + \frac{2!}{2^2} + \dots + \frac{N!}{2^N}$$

- a)

```
total = factorial (N) / power(2, N);
```
- b)

```
total = 0;
for i = 1 : N
    total = total + factorial (i) / power(2, N);
end
```
- c)

```
total = 0;
for i = 1 : N
    total = total + factorial (N) / power(2, i);
end
```
- d)

```
total = 0;
for i = 1 : N
    total = total + factorial (i) / power(2, i);
end
```
- e)

```
total = 0;
for i = 1 : N
    total = total + factorial (N) / power(2, N);
end
```

7. Which of the following is not included in the function prototype?

- a) Name of the method
b) List of the parameters
b) Name of the parameters
b) "function" string
e) Contents of the method

8. Let say you have the function factorial

```
function result = factorial (N)
```

which calculates $N!$. Which of the following code fragments then calculate the following:

$$P = \frac{N!}{(N - M)!}$$

- a)

```
P = factorial (N, M);
```
- b)

```
P = 0;
for i = 1 : N
    P = P + factorial (N, M);
end
```

c)

```
P = factorial (N) / factorial (N - M);
```

d)

```
P = 0;
for i = 1 : N
    P = P + factorial (N - M, i);
end
```

e)

```
P = 0;
for i = 1 : M
    P = P + factorial (N, i);
end
```

9. Let say you have the function

```
function result = sumOfDivisorsExceptItself(N)
```

which returns the sum of divisors of N except itself. Which of the following code fragments prints all perfect numbers until 1000, where a number is perfect if it is equal to the sum of its divisors?

a)

```
disp(sumOfDivisorsExceptItself(100));
```

b)

```
for i = 2 : 1000
    disp(sumOfDivisorsExceptItself(i));
end
```

c)

```
for i = 2 : 1000
    if sumOfDivisorsExceptItself(i) == true
        disp(i);
    end
end
```

d)

```
for i = 2 : 1000
    disp(i);
end
```

e)

```
for i = 2 : 1000
    if sumOfDivisorsExceptItself(i) == i
        disp(i);
    end
end
```

10. What is the content of the array list after executing the following code?

```
myList = [10, 20, 30, 40];
tmp = myList(length(myList));
for i = 2 : length(myList)
    myList(i) = myList(i - 1);
end
myList(1) = tmp;
```

- a) {10, 20, 30, 40}
- b) {40, 10, 10, 10}
- c) {40, 40, 40, 10}
- d) {10, 40, 40, 40}
- e) {10, 10, 10, 10}

11. Which of the following functions takes a parameter of an array and a number N, and returns the number of elements of that array that are smaller than N?

a)

```
function smaller(array, N)
    N = input();
    count = 0;
    for i = 1 : length(array)
        if array[i] < N
            count = count + 1;
        end
    end
end
```

b)

```
function smaller(array, N)
    count = 0;
    for i = 1 : length(array)
        if array[i] < N
            count = count + 1;
        end
    end
end
```

c)

```
function count = smaller(array, N)
    count = 0;
    for i = 1 : length(array)
        if array[i] < N
            count = count + 1;
        end
    end
end
```

d)

```
function count = smaller(array)
    count = 0;
    for i = 1 : length(array)
        if array[i] < N
            count = count + 1;
        end
    end
end
```

e)

```
function count = smaller(array, N)
    count = 0;
    for i = 1 : length(array)
        if array[i] > N
            count = count + 1;
        end
    end
end
```

end

12. Assume that we have a method which calculates the sum of the elements in an array. Assume that method has a prototype **function result = findSum(arr)**. Which one is the correct way to call this method?

- a) findSum(arr[]);
- b) sum = findSum(arr);
- c) sum = findSum(arr[length(arr)]);
- d) findSum(int arr[]);
- e) None of the above

13. Which of the following can be used to get the last element of an array?

- a) arr(arr.length)
- b) arr(arr.length-1)
- c) arr(arr.size)
- d) arr(length(arr))
- e) arr(arr.size())

14. Which of the following functions takes a parameter of an array, and returns the average of the elements of that array?

a)

```
function averageArray(array)
    average(array);
end
```

b)

```
function averageArray(array)
    average = 0;
    for i = 1 : length(array)
        average = average + array(i);
    end
end
```

c)

```
function averageArray(array)
    average = 0;
    for i = 1 : length(array)
        average = average + array(i);
    end
    average = average / length(array);
end
```

d)

```
function result = averageArray(array)
    average = 0;
    for i = 1 : length(array)
        average = average + array(i);
    end
    average = average / length(array);
end
```

e)

```
function average = averageArray(array)
    average = 0;
```

```

for i = 1 : length(array)
    average = average + array(i);
end
average = average / length(array);
end

```

15. Which of the following is the MATLAB counterpart for the following math expression?

$$\frac{x^2 + 2}{3y - 5}$$

- a) $(x * x + 2) / (3y - 5)$
- b) $x * x + 2 / 3 * y - 5$
- c) $(x * x + 2) / (3 * y - 5)$
- d) $x^2 + 2 / (3 * y - 5)$
- e) $(x * x + 2) / 3y - 5$